CHEST Critical Care

Investigation of Skin Pigmentation Effect on Performance of Pulse Oximetry

Feasibility of a Study Examining the Effect of Skin Pigment on Pulse Oximetry

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BACKGROUND: The accuracy of blood oxygen saturation measured by pulse oximetry (Sp_{O_2}) in critically ill patients with darkly pigmented skin remains controversial, with accuracy studies that are either retrospective or laboratory based among healthy volunteers.

RESEARCH QUESTION: Can a pragmatically-designed prospective study feasibly enroll critically ill patients across the full skin pigmentation spectrum within a real-world ICU setting while applying the technical rigor of a laboratory-based study?

STUDY DESIGN AND METHODS: A feasibility analysis was performed after 50 patients across 2 ICUs contributed samples. Feasibility was defined as (1) coverage of skin pigmentation, with a range of $> 100°$ using an objective measure (individual topology angle [ITA]) and (2) $\geq$ 80% of collected blood samples generate a valid Sp_{O_2} and Sa_{O_2} pair fulfilling predefined technical criteria. Exploratory analysis was performed through Bland-Altman analysis, with standard-of-care blood draws used for Sa_{O_2} measurements. Rates of occult hypoxemia ($Sa_{O_2} < 88\%$ when Sp_{O_2} was 92%-96%) were compared among skin pigmentation groups.

RESULTS: Eighty-four blood samples collected from 50 patients (52% male; median age, 55 years [interquartile range, 46-61.8 years]) yielded 80 valid pairs (95%) from 49 patients. Feasibility thresholds were met for skin pigmentation (ITA range, –66° to 59°), with 52% of patients classified as dark-skinned overall. In the exploratory analysis, overall pulse oximeter root-mean-square accuracy was 1.47%, with a median bias of –0.6% (mean, –0.7%). Dark-skinned and light-skinned patients did not differ significantly, with a median bias of –0.2% vs –0.9% (mean, –0.1% vs –1.1%), respectively. Nearly one-half the data pairs were collected when patients were at risk of occult hypoxemia (Sp_{O_2}, 92%-96%), yet no occult hypoxemic events occurred.

INTERPRETATION: This feasibility study demonstrated high fidelity and efficient data collection from pulse oximeter sensors and reference blood gas analyzers in critically ill patients across varying skin pigments while applying requisite technical criteria within an ICU setting.

CLINICAL TRIAL REGISTRATION: ClinicalTrials.gov; No.: NCT06432881; URL: www.clinicaltrials.gov

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KEY WORDS: bias; ethnicity; occult hypoxemia; oxygen saturation; race

Take-Home Points

Study Question: What is the feasibility of a prospective study design to evaluate pulse oximetry performance and the frequency of occult hypoxemia accurately in patients with light and dark skin pigmentation in a real-world ICU setting using the technical rigor of a laboratory-based study?

Results: The study design met the enrollment feasibility threshold while applying the requisite technical criteria for blood oxygen saturation measured by pulse oximetry and arterial blood gas oxygen saturation sampling; nearly one-half of the data pairs were collected when patients were at risk of occult hypoxemia, yet no occult hypoxemic events were observed.

Interpretation: These feasibility data indicate that the study design enables efficient enrollment of critically ill patients across the full spectrum of skin pigmentation and facilitates accurate data collection in a real-world ICU setting.

Pulse oximetry is a ubiquitous technology used to monitor arterial blood oxygenation in patients at risk of hypoxemia. Despite its widespread use, laboratory investigations have demonstrated differences in the accuracy of some pulse oximetry sensors across skin pigmentation in people with darker skin pigmentation over the past several decades.[1-3] Further, clinical cohort studies have suggested similar disparities in its accuracy among Black patients, resulting in disproportionate rates of occult hypoxemia, defined as low arterial blood gas (ABG) oxygen saturation (Sa_{O_2}; $<$ 88%) despite acceptable blood oxygen saturation measured by pulse oximetry (Sp_{O_2}; 92%-96%).[4-7] Multiple large cohort studies have demonstrated rates of occult hypoxemia for Black patients as high as 5% to 11%, suggesting a clinically meaningful risk.[4-7] However, these studies were limited by the use of self-reported race as a surrogate for skin pigmentation and retrospective designs that precluded assessment of the plethysmography waveform quality and stability and resulted in prolonged lag times between Sp_{O_2} and Sa_{O_2} readings.[4-8] Additionally, studies have failed to specify device manufacturer, model, and sensor location. Collectively, these limitations inhibit the ability to make conclusions about the impact of skin pigmentation on device accuracy.

Pulse oximetry accuracy studies conducted in the laboratory setting apply the technical criteria for precise data collection necessary to assess device accuracy.[9] However, controlled laboratory desaturation protocols are limited to healthy volunteers and do not reflect clinical conditions that can confound pulse oximetry accuracy, including hypoperfusion, iatrogenic vasoconstriction, motion artifact, anemia, and hyperbilirubinemia.[10,11] Currently, the combined impact of low perfusion states and skin pigmentation on pulse oximetry accuracy in the ICU is unknown. Accordingly, the US Food and Drug Administration (FDA) has urged further investigations using objective skin pigmentation assessments and temporally correlated measurements in a real-world setting.[12] However, failure to apply the technical principles for accurate Sp_{O_2} and Sa_{O_2} paired data collection can impact data integrity, thereby limiting the generalizability of results.

To address this unmet need, we designed the Investigation of Skin Pigmentation Effect on Performance of Pulse Oximetry (INSPIRE) study to integrate the technical rigor of a laboratory study into the ICU setting. Essential considerations in the study design included: (1) the need to compare simultaneous peripheral (Sp_{O_2}) with blood reference (Sa_{O_2}) arterial saturations, (2) the ability to quantify skin pigmentation objectively with adequate coverage of skin pigmentation range, (3) investigating disparities in the outcome of occult hypoxemia, and (4) optimizing efficient enrollment and data capture. Further, the study was designed to examine patients with the greatest need for accurate monitoring and at the greatest risk of confounding. We elected to test a pulse oximeter device

ABBREVIATIONS: ABG = arterial blood gas; A_{RMS} = root-mean-square accuracy; FDA = US Food and Drug Administration; INSPIRE = Investigation of Skin Pigmentation Effect on Performance of Pulse Oximetry; ITA = individual topology angle; Sa_{O_2} = arterial blood gas oxygen saturation; Sp_{O_2} = blood oxygen saturation measured by pulse oximetry

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and sensor (Masimo RD SET) demonstrated to measure Sp_{O_2} accurately in both dark-skinned and light-skinned healthy participants,[13] including those with normal and low perfusion indexes.[14]

Prospective studies integrating the technical rigor of laboratory Sp_{O_2} and Sa_{O_2} sampling methods in critical care settings are not well established. Thus in this study, we aimed to investigate the feasibility of the INSPIRE design to efficiently enroll critically ill patients who are broadly representative of the full spectrum of skin pigmentation and to optimize high-fidelity data collection within a real-world critical care setting. An exploratory analysis evaluated the application of requisite Sp_{O_2} and Sa_{O_2} sampling techniques for accuracy assessment among patients at risk of occult hypoxemia under standard-of-care workflows.

Study Design and Methods

Trial Design

This feasibility study of the prospective, observational INSPIRE trial enrolled adult patients admitted to the ICU with hypoxemia defined by (1) the receipt of supplemental oxygen or (2) peripheral oxygenation of ≤ 94% on room air. Using a pragmatic approach, the study enrolled eligible participants in a sequential fashion using a waiver of informed consent. The study protocol (Identifier: Pro00129448) was approved by the institutional review board at the Medical University of South Carolina, and this trial was registered with ClinicalTrials.gov (Identifier: NCT06432881) in compliance with US Food and Drug Administration Amendents Act section 801 (e-Table 1). At any time during study procedures, the treating physician and clinical team could discontinue study procedures and exercise clinical judgment to safeguard the patient's health, safety, and welfare.

Study Population

Feasibility study enrollment occurred at 2 ICUs located within Medical University of South Carolina Health University Medical Center from January through July 2024. Patients were eligible for inclusion if they were an adult, admitted to a participating ICU, met the study definition of hypoxemia, and had an upper extremity arterial catheter in place. The arterial catheter enabled the use of convenient sampling from standard-of-care ABG analyses, thereby eliminating the need for study-specific sample collection. Patients were excluded if pulse oximetry could not be performed reliably on the second, third, or fourth digit of the ipsilateral hand or had only lower extremity arterial catheters in place. A complete list of inclusion and exclusion criteria is provided in e-Table 2.

Enrollment: Feasibility Study Cohort

Of the first 107 enrolled patients, 50 patients yielded at least 1 blood sample for the study, whereas the remaining 57 patients did not have a clinically indicated ABG collected while the study team was present. This feasibility sample size was chosen because it represents nearly 10% of the anticipated INSPIRE trial sample size of 502, assuming a single sample per patient. As such, this population would allow for an early assessment of study feasibility while allowing for timely protocol amendments if required.

Skin Pigmentation Assessment

To account for the limitation of self-reported race as the sole determinant of skin pigmentation used in prior studies,[4-8] this study used an objective measure of skin pigmentation, the individual topology angle (ITA).[15] The ITA of enrolled patients was measured using the SkinColorCatch device (Delfin USA), a validated, noninvasive handheld skin colorimeter. Patient skin pigmentation was dichotomized into light skinned, which includes the validated ITA categories very light, light, intermediate, and tan (ITA, 10°-90°), and dark skinned, which includes ITA categories brown and dark (ITA, 9° to −90°).[15,16] To improve measurement accuracy, the median of 3 measurements obtained on the forehead from a dry, smooth region free of hair, scars, folds, and hypopigmentation and hyperpigmentation was used to designate a patient's skin pigmentation consistent with FDA guidance.[17]

Sensor Configuration and Data Acquisition

Patients continued to be monitored by standard-of-care devices while enrolled in this study. In addition, a Masimo RD SET sensor was placed on the patient's second, third, or fourth digit, ipsilateral to the arterial line and contralateral to the BP cuff. To mitigate extraneous optical interference (ie, crosstalk) from the standard-of-care pulse oximeter, an optical shield was placed over the study sensor.[18] The clinical team notified the research team in advance of a clinically indicated ABG such that the research team could connect the study probe to the Masimo Rad-97 Pulse CO-Oximeter before blood collection. The study monitor

Signal IQ was assessed for adequate signal intensity, and only samples collected in the absence of a low Signal IQ warning were included in the study. Automated data collection software was used to capture the exact time of the blood draw, enabling comparison with the pulse oximetry values recorded at the exact time of blood gas collection. Samples were collected for analysis only if the standard-of-care Sp_{O_2} was observed by the study team as stable at the time of the ABG collection, with stability defined as no change in value more than ± 1% for the 30 seconds before ABG collection. Samples collected within 24 hours of a noncontrast IV dye such as methylene blue were not captured for analysis.[19]

As soon as the blood sample was drawn, air bubbles were removed promptly and the clinical team applied some of the blood to the point-of-care ABG analyzer. The remaining waste blood in the syringe was handed to the study team member, who immediately analyzed it via co-oximetry (ABL 90 FLEX; Radiometer) before disposing of residual blood. The peripheral saturation measurements from the study sensor were downloaded regularly, and the Sp_{O_2} value corresponding to the time of blood collection was used to create a time-matched Sp_{O_2} and Sa_{O_2} pair.

The frequency and number of samples could vary between patients. To enrich the sample pool with specimens at risk of occult hypoxemia, a patient could not contribute > 2 samples unless the Sp_{O_2} was ≤ 96% as an evidence-based threshold for individuals at risk of occult hypoxemia at darker skin tones.[4-7] Relevant clinical data that could influence the accuracy of pulse oximetry were captured at the time of ABG sample collection including: BP, vasopressor requirements, conditions that impact fingertip perfusion (eg, mottling, Raynaud's phenomenon, peripheral vascular disease), dyshemoglobinemias, hemoglobinopathies, bilirubinemia, and recent receipt of other vasoactive agents (e-Table 3).

Outcomes

In this feasibility study, end points included the distribution of patients stratified by skin pigmentation measured by forehead ITA. The threshold for feasibility was enrollment of patients with skin pigmentation that covered a wide range of ITA measurements (> 100°) and represented each of the 6 ITA categories for light (very light, light, intermediate, and tan [ITA, 10°-90°]) and dark (brown and dark [ITA, 9° to –90°]) skin. This feasibility threshold was selected to reflect the latest FDA draft guidance for pulse oximetry performance testing.[17] Technical feasibility end points included fulfillment of Sp_{O_2} and Sa_{O_2} paired sampling methods including appropriate sensor placement, time synchronization, and adequate signal quality with exclusion of Sp_{O_2} collected when a low Signal IQ warning present.[8] Technical feasibility was defined as at least 80% of captured standard-of-care blood draws contributing paired Sp_{O_2} and Sa_{O_2} samples meeting all study criteria. This acceptance criterion was chosen to align with the study protocol design, which was powered with the assumption that up to 20% of samples would not meet the aforementioned technical standards.

For the exploratory analysis, data collected from the feasibility cohort were used to calculate the root-mean-square accuracy (A_{RMS}) of the study sensor in patients with light and dark skin using the arterial saturation measured by co-oximetry as the reference value. This included the proportion of patients with light and dark skin pigmentation who experienced occult hypoxemia, defined as an Sa_{O_2} of < 88% despite an Sp_{O_2} of 92% to 96% (e-Table 4).[4,5] This patient-centered outcome was selected based on clinical relevance and inclusive of individuals sufficiently at risk for occult hypoxemia (ie, reported prevalence, ≥ 5%).[7]

Statistical Analysis

The sample size of the INSIPIRE trial was estimated using constants reported in the 2020 correspondence by Sjoding et al.[4] The prevalence of occult hypoxemia was set to 0.05 for the light pigment group and 0.14 for the dark pigment group. The two one-sided tests equivalence margin was set to the difference between light and dark averages (0.09). Using 2 1-sided Z tests and assuming a significance level of $P < .05$, a sample size of 251 people per skin pigmentation group (502 people total) was estimated to achieve 82% power to detect equivalence. Because this sample was substantially larger than the sample required to address the A_{RMS} outcome, this sample was used to power the overall study.

Descriptive statistics were used to report patient demographics and clinical characteristics of the feasibility study cohort using counts and percentages for categorical data. To assess enrollment feasibility, counts were used to report the distribution of patients stratified by skin pigmentation measured by forehead ITA in degrees. Technical feasibility end points were reported as the proportion of captured standard-of-care blood draws contributing paired Sp_{O_2} and Sa_{O_2} samples meeting all technical criteria.

For the exploratory analysis, Bland-Altman statistics were used to compare pulse oximetry readings with co-oximetry measures. The bias, precision (SD of the difference), and A_{RMS} of patients were calculated for each of the 2 pigmentation groups. The Kolmogorov-Smirnov test for normality was used to determine if nonparametric methods are required and the analysis of variance or Kruskal-Wallis test was used to compare the biases between skin pigmentation groups. The Bland-Altman and comparison analyses initially were performed by the study sponsor and were confirmed independently by the study investigators. The rate of occult hypoxemia among light and dark skin pigmentation groups was reported.

Results

Participants in the Feasibility Study Cohort

Descriptive characteristics of the 50 patients are presented in Table 1. Of these 50 patients, 26 patients had dark-pigmented skin, nearly all of whom self-identified as Black. Pneumonia was the most common primary diagnosis of the feasibility study cohort, whereas diabetes mellitus and congestive heart failure were the most common predefined comorbid conditions that may impact peripheral perfusion.

Feasibility

The feasibility cohort (Fig 1) met the threshold for enrollment feasibility, because the skin pigmentation of enrolled patients covered a wide range of ITA ($> 100°$)[17] and included each of the 6 skin color subtypes[15] defined by ITA values. For the technical feasibility end points, 80 Sp_{O_2} and Sa_{O_2} paired samples met all technical criteria of 84 standard-of-care blood draws (95%) obtained in the study (Table 2).

TABLE 1] Baseline Demographics, Admitting Diagnoses, and Comorbidities Stratified by Skin Pigmentation

Variable	All Patients (N = 50)	Dark-Skinned Patients (n = 26)	Light-Skinned Patients (n = 24)
Age, y	55 (46-61.8)	57 (46.3-62.8)	53 (45.2-60.3)
Female sex	24 (48)	13 (50)	11 (45.8)
Race			
Black	26 (52)	25 (96.2)	1 (4.2)
White	23 (46)	1 (3.8)	22 (91.7)
Other[a]	1 (2)	0 (0)	1 (4.2)
Ethnicity			
Hispanic	0 (0)	0 (0)	0 (0)
Non-Hispanic	50 (100)	26 (100)	24 (100)
Cause of hypoxemia			
Pneumonia	24 (48)	13 (50)	11 (45.8)
Other respiratory failure	17 (34)	9 (34.6)	8 (32)
Cardiac arrest	3 (6)	1 (3.8)	2 (8)
Encephalopathy	6 (12)	3 (11.5)	3 (12)
Baseline comorbid conditions			
Cerebrovascular disease	1 (2)	0 (0)	1 (4.2)
Congestive heart failure	6 (12)	5 (19.2)	1 (4.2)
Diabetes mellitus	7 (14)	6 (23)	1 (4.2)
Coronary artery disease	1 (2)	0 (0)	1 (4.2)
Peripheral vascular disease	1 (2)	0 (0)	1 (4.2)
Raynaud's syndrome	1 (2)	0 (0)	1 (4.2)
Oxygen support type			
Ventilator	41 (82)	21 (80.8)	20 (83.3)
High-flow nasal cannula	7 (14)	3 (11.5)	4 (16.7)
Nasal cannula	2 (4)	2 (7.7)	0 (0)

Data are presented as No. (%) or median (interquartile range).
[a]Indicates any race other than Black or White.

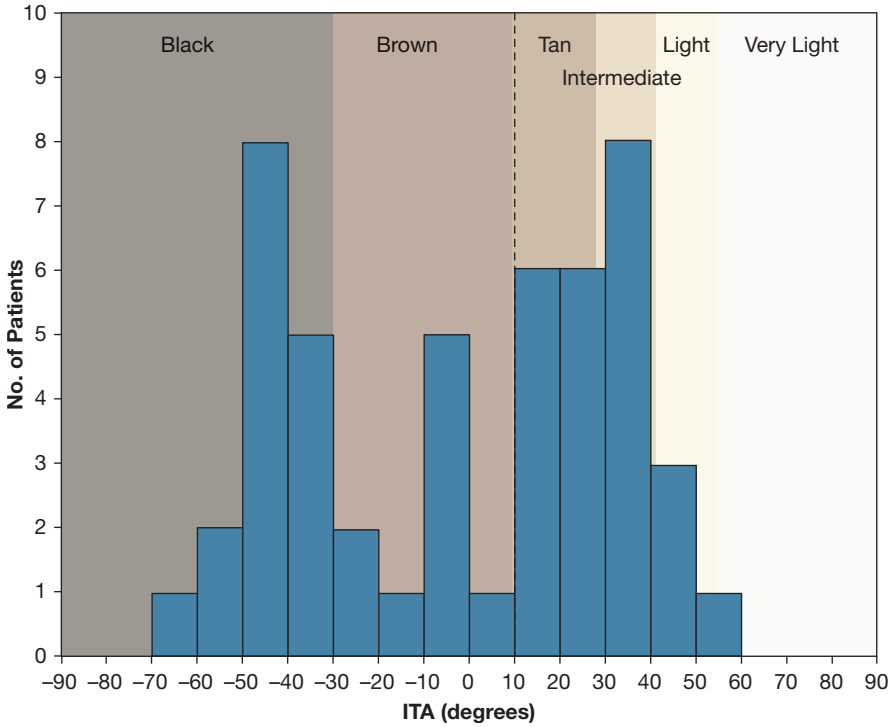

Figure 1 – *Bar graph showing the distribution of patients stratified by skin pigmentation measured by forehead ITA (in degrees): black, < –30°; brown, –30° to 10°; tan, 10°-28°; intermediate, 28°-41°; light, 41°-55°; and very light, 55°-90°. The vertical dashed line (ITA, 10°) indicates the cutoff used in the study for separation of light and dark pigment subgroups. ITA = individual topology angle.*

Exploratory Analysis

Forty-nine patients contributed the 80 ABGs that met all technical criteria. Most patients were receiving vasopressors at the time of the ABG sample collection (65%), and many patients showed elevated bilirubin levels (35%) (Table 3). Thirty-one data pairs (39%) were collected in the setting of a low (< 1.4%) perfusion index (Fig 2).

The mean pulse oximeter bias of this feasibility data set was –0.68% with an A_{RMS} of 1.47% (Fig 3). Separate assessment of each pigment group revealed a nonnormal distribution of the error among lighter-pigmented patients. A small but statistically significant difference of 0.7% was found in the median bias between the dark-pigmented and light-pigmented groups (–0.2% vs –0.9%, respectively; $P = .0004$) (Fig 4). Although nearly one-half of the patients were in the occult hypoxemia risk zone (Sp_{O_2}, 92%-96%), no occult hypoxemic events were observed in either skin pigment group (Table 3).

TABLE 2] Relevant Technical Criteria Applied to Paired Sp_{O_2} and Sa_{O_2} Samples Within Standard-of-Care Workflows in the ICU Care Setting

Variable	Study Samples (N = 84)
Proper Sp_{O_2} sensor placement	84 (100)
Adequate signal quality of Sp_{O_2} monitor	81 (96)
Time synchronization of Sp_{O_2} and Sa_{O_2} sampling	84 (100)
Other exclusions (no longer met hypoxemia criteria)	1 (1)
Qualified pairs	80 (95)

Data are presented as No. (%). Sa_{O_2} = arterial blood gas oxygen saturation; Sp_{O_2} = blood oxygen saturation measured by pulse oximetry.

Discussion

In this study, we demonstrated that the INSPIRE trial design feasibly can enroll critically ill patients across the full spectrum of skin pigmentation while facilitating precise data collection using standard-of-care workflows in an ICU setting. Skin pigmentation of the cohort covered an ITA range of > 100° and represented all 6 skin color subtypes, whereas the data collection methods met the predefined technical criteria. Moreover, the exploratory analysis demonstrated that the rigorous Sp_{O_2} and Sa_{O_2} sampling criteria can be applied in a dynamic ICU setting to capture high-fidelity data for the evaluation of pulse oximetry accuracy among critically ill patients at risk of occult hypoxemia.

Recent studies have suggested that pulse oximeters may be inaccurate in acutely ill patients with dark skin pigmentation and that occult hypoxemia may be

TABLE 3] Paired Pulse Oximetry and ABG Outcomes by Skin Pigment Group

Variable	All Study ABG Samples (N = 80)	Dark-Skinned ABG Samples (n = 41)	Light-Skinned ABG Samples (n = 39)
Spo_2			
< 90%	6 (7.5%)	4 (9.8%)	2 (5.1%)
90%-96%	37 (46.3%)	17 (41.5%)	20 (51.3%)
> 96%	37 (46.3%)	20 (48.8%)	17 (43.6%)
Bias			
Median (IQR)	−0.6 (−1.5 to 0.1)	−0.2 (−1.1 to 0.6)	−0.9 (−1.8 to −0.3)
Mean (SD)	−0.7 (1.3)	−0.1 (1.1)	−1.2 (1.3)
Occult hypoxemia	0 (0%)	0 (0%)	0 (0%)
Mean arterial pressure	75.8 (70.2-83)	73.3 (69.8-81.3)	79.3 (73.4-85.3)
Vasopressor use	52 (65%)	27 (65.9%)	25 (64.1%)
Other vasoactive drugs[a]	11 (13.8%)	6 (14.6%)	5 (12.8%)
Bilirubin elevation[b]	28 (35%)	10 (24.4%)	18 (46.2%)

Data are presented as No. (%), mean (SD), or median (interquartile range). ABG = arterial blood gas.
[a]Vasoactive drugs administered < 24 hours before ABG sampling.
[b]Normal total bilirubin range, 0.2-1.2 mg/dL.

harmful because recent data suggest an association with all-cause mortality.[20] However, our ability to characterize the validity and magnitude of this phenomenon definitively has been limited by retrospective study designs, use of calculated readings as gold standards, imprecise methods of skin pigmentation categorization, and an inability to control for confounders.[4-8,21] The INSPIRE study was designed to

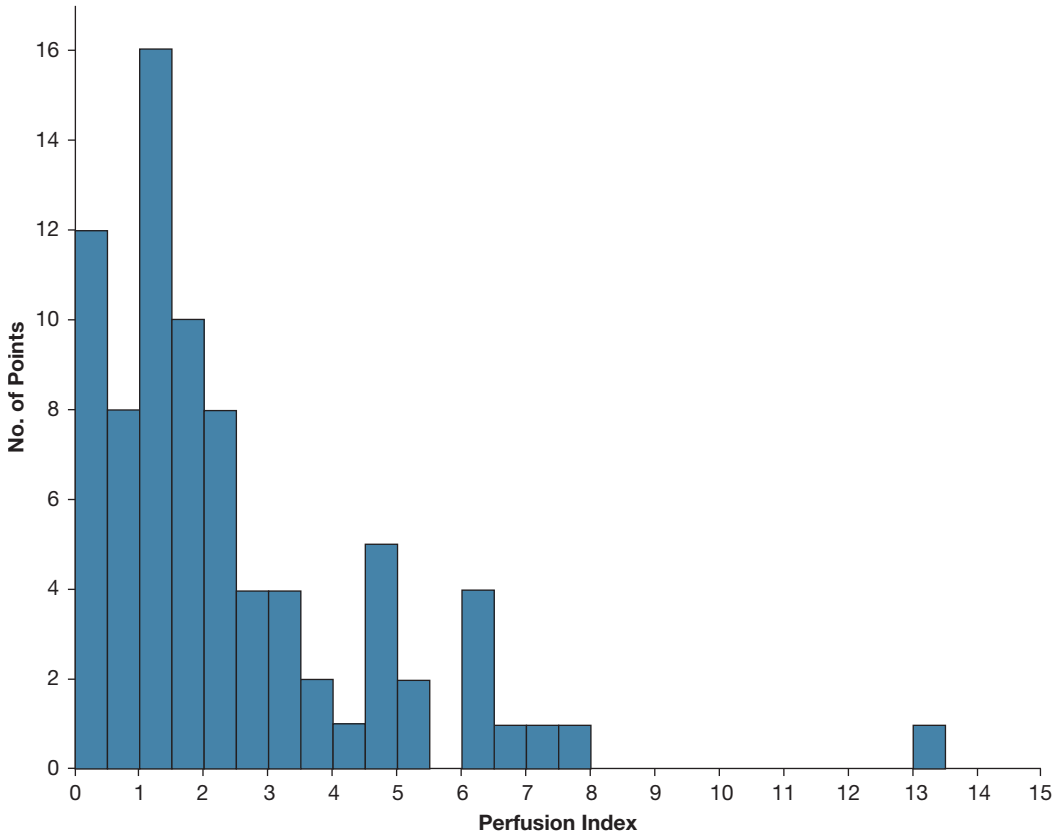

Figure 2 – *Bar graph showing the distribution of perfusion index for each individual arterial blood gas sample.*

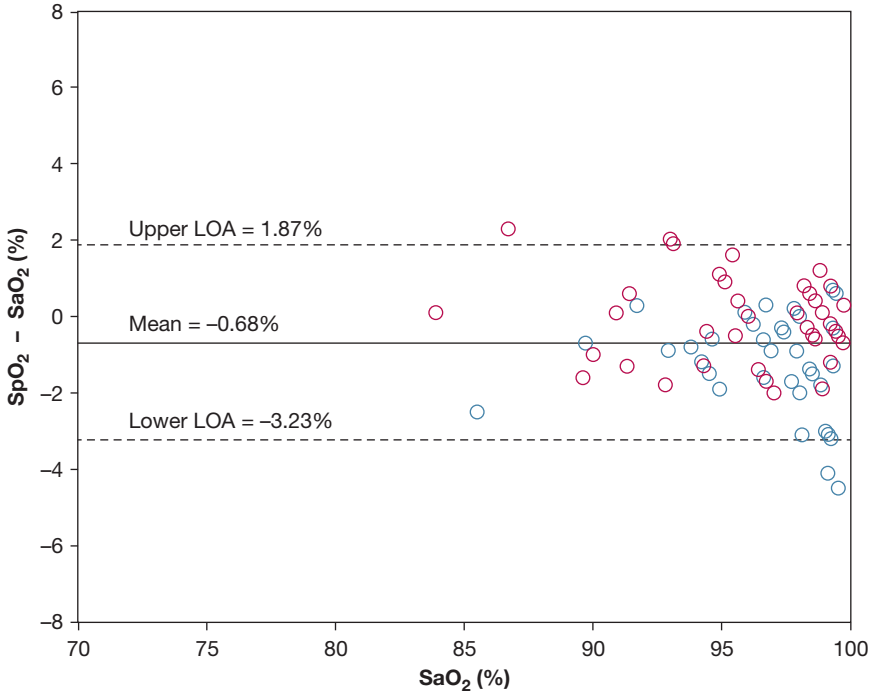

Figure 3 – *Graph showing the distribution of pulse oximeter bias per arterial blood gas sample by skin pigmentation. N = 80; bias = –0.68%. Upper LOA = 1.87%; lower LOA = –3.23%. Blue circle indicates light-skinned group; brown circle indicates dark-skinned group. LOA = limit of agreement; Sa_{O_2} = arterial blood gas oxygen saturation; Sp_{O_2} = blood oxygen saturation measured by pulse oximetry.*

overcome these challenges while examining the clinically relevant outcome of occult hypoxemia in a population vulnerable to this diagnostic error. The feasibility results support the present study design to address this unmet need among a larger cohort of critically ill patients. The study's pragmatic design features efficient patient enrollment across the spectrum of skin pigmentation broadly at high risk of occult hypoxemia. Salient cohort features include high rates of potentially confounding features including shock (ie, high vasopressor requirement), relevant comorbidities, low peripheral perfusion, and hyperbilirubinemia.[1,22-24]

Skin pigmentation can vary widely across racial and ethnic groups, and subjective assessments often correlate poorly with actual skin tone.[25] The latest FDA

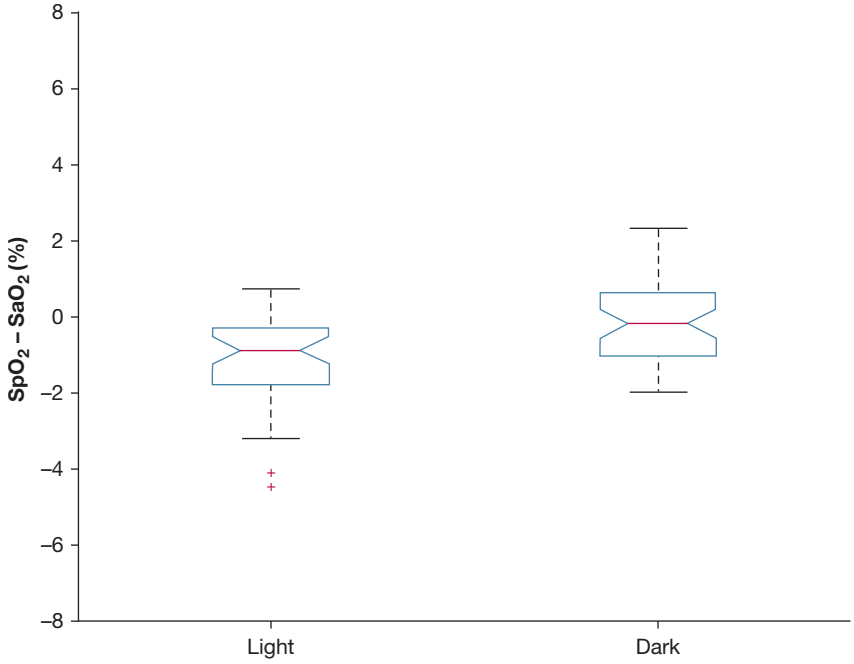

Figure 4 – *Graph showing the distribution of bias by skin pigmentation. Median bias for light pigment individuals, –0.9%; median bias for dark skin, –0.2% (P = .00004). Sa_{O_2} = arterial blood gas oxygen saturation; Sp_{O_2} = blood oxygen saturation measured by pulse oximetry.*

draft guidance for pulse oximeter performance testing recommends measurement of patient forehead pigmentation using colorimetry to determine the ITA.[17] The Delfin SkinColorCatch device is a validated tool used to measure skin tone score objectively for each patient.[26] Using a consecutive enrollment strategy with pragmatic data collection, this feasibility sampling included patients with a broad range of skin tones to include the full spectrum. These preliminary observations combined with a planned large sample size support the feasibility of this study not only to compare the accuracy of the study sensor across a dichotomous classification of light and dark skin pigmentation, but also to examine its accuracy across the entire spectrum of skin pigmentation consistent with FDA recommendations.[17]

Prospective studies examining the impact of skin pigmentation on pulse oximetry performance in critically ill patients have been limited to relatively small sample sizes with methodologic constraints.[27,28] This study adds to the literature by piloting a prospective design that integrates the technical principles of a laboratory study with standard-of-care procedures in a complex, high-acuity clinical setting. Sp_{O_2} data are dynamic and prone to artifact, which can be amplified by data collection errors. Incorporating the essential technical criteria for Sp_{O_2} and Sa_{O_2} sampling mitigates measurement inaccuracies to generate results that appropriately reflect device performance. The present analysis demonstrates technical and clinical feasibility to use standard-of-care blood draws in the ICU, thereby eliminating the need for study-specific blood sample collection.

The exploratory analysis demonstrated that the applied Sp_{O_2} and Sa_{O_2} sampling techniques generate precise data that can be used to evaluate pulse oximeter accuracy among critically ill patients at risk of occult hypoxemic events. The data showed that most (54%) ABGs captured during this feasibility sampling were drawn when the patient showed an Sp_{O_2} of $\leq$ 96%. This is a substantially higher proportion of at-risk patients than reported in a contemporary study also investigating this topic.[28] This difference in proportions may increase further as enrollment progresses because of INSPIRE's design, which enriches for the collection for ABGs at lower saturations. Results of the exploratory analysis suggest a modest difference in measurement bias (Sp_{O_2} and Sa_{O_2}) between the 2 skin pigmentation groups, with less bias seen in the dark skin pigmentation group. The exploratory results are consistent with laboratory study results obtained from similar devices[13,14]; however, the final study data set must be analyzed before more definitive conclusions can be drawn.

This feasibility study demonstrated patient enrollment across the full spectrum of skin pigmentation with a wide variety of clinical conditions. As such, the results of the completed study will provide insights into the accuracy of pulse oximetry among a wide range of patients at risk of measurement error. An important limitation to these insights is that they are confined only to the sensor being examined in this study. Additionally, some ABGs were collected as part of clinical care from enrolled patients that were not captured for the study, because 24-hour staffing of research personnel in the ICU was not available. However, we do not believe that this sampling approach is imparting bias into the data set because no known link exists between the time of day and the association of skin pigmentation with Sp_{O_2} accuracy. Finally, the INSPIRE study will include the comparison of occult hypoxemia rates between patients with light and dark pigmentation as a primary outcome to ensure that the study will be powered to address this clinically important finding.

Interpretation

This study design facilitated accurate and efficient data collection from pulse oximeter sensors and ABG measurements in critically ill patients with varying skin pigments. These data support the study's feasibility in a real-world clinical setting and the appropriateness of the chosen methods.

Funding/Support

Masimo Corp. (Irvine, California) provided the RD SET sensors, the Rad-97 pulse oximeters, and the Radiometer ABL90 co-oximeter and supplies for the study as well as additional funding for study performance.

Financial/Nonfinancial Disclosures

None declared.

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Role of sponsors: The sponsor provided funding, codesigned the study with the investigators, and performed the initial statistical analysis, which was repeated and verified by the investigators. The sponsor had no involvement in data collection. The sponsor reviewed the manuscript for scientific accuracy and clarity, but did not have any influence on the interpretation of the data.

Additional information: The e-Tables are available online under "Supplementary Data."

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